

A pure exchange economy with 2 consumers (1 and 2) and 2 goods (x and y):

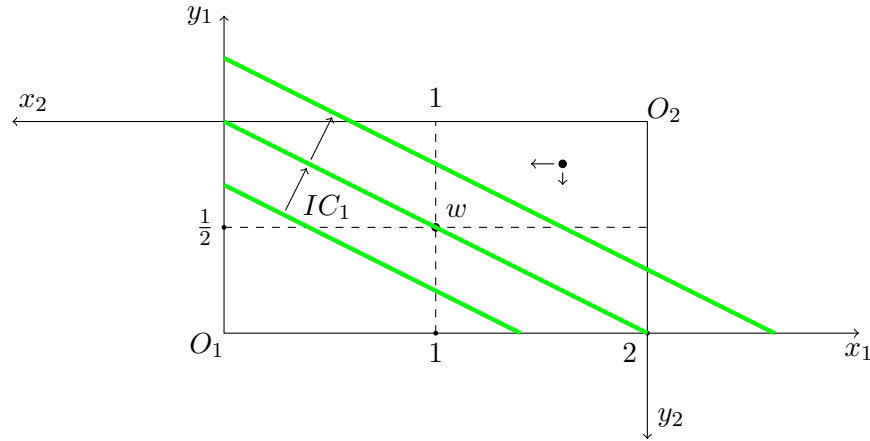
consumption spaces are $X_i = \mathbb{R}_+^2$ for $i = 1, 2$;

each consumer's **initial endowment** is $w_i = (1, \frac{1}{2})$ for $i = 1, 2$;

consumer 2's preferences are lexicographic with x the first priority.

consumer 1's utility function is $u_1(x_1, y_1) = x_1 + 2y_1$.

(a) **Draw a diagram representing the above economy.**



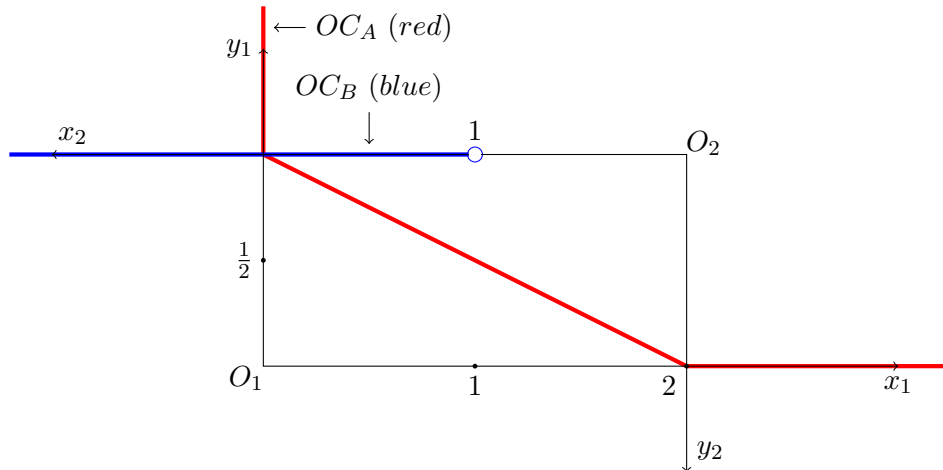
(b) **Fund demand functions for each consumer.**

Let $p^y/p^x = p^1$, then the demand function/correspondence for agents are:

$$(x_1(p), y_1(p)) = \begin{cases} (0, \frac{1}{2} + \frac{1}{p}), & \text{if } 0 < p < 2; \\ \{(x, y) \in \mathbb{R}_+^2 : x + 2y = 2\}, & \text{if } p = 2; \\ (1 + \frac{p}{2}, 0), & \text{if } p > 2. \end{cases}$$

$$(x_2(p), y_2(p)) = (1 + \frac{p}{2}, 0).$$

(c) **Draw the offer curves in the Edgeworth box diagram.**



¹Clearly, if $p^x = 0$ or $p^y = 0$, at least one consumer would demand infinite quantity of x or y , and thus market clearing condition cannot hold. Therefore, we do not consider zero price in this economy.

(d) **Find the competitive equilibrium if any. If not, consider the reason why.**

The only CE is $\{(x_1, y_1), (x_2, y_2), (p^x, p^y)\} = \{(0, 1), (2, 0), (1, 2)\}$, which can be found at the intersection of offer curves. We can also find the CE by market clearing condition. $\square$

Proposition 7.4.1: For a private ownership economy $e = (\{X_i, w_i, \succsim_i\}, \{Y_j\}, \{\theta_{ij}\})$, under local non-satiation assumption, we have the budget constraint holds with equality, and thus the **Walras's Law** holds. That is $p[\sum_{i=1}^n x_i - (\sum_{j=1}^J y_j + \sum_{i=1}^n w_i)] = 0$.

An allocation (x, y) is **balanced** if $\sum_{i=1}^n x_i = \sum_{j=1}^J y_j + \sum_{i=1}^n w_i$.

Suppose Walrasian equilibrium exists for an economy.

1). Is the equilibrium allocation balanced if preferences satisfy **local non-satiation**?

2). Is the equilibrium allocation balanced if preferences satisfy **monotonicity**?

3). Is the equilibrium allocation balanced if preferences satisfy **strict monotonicity**?

Only the answer for 3) is yes. We may find counter-example of 1) and 2) in our homework.

Proof of 3): strict monotonicity implies local non-satiation. By the First welfare theorem of chapter 8, under local non-satiation, every CE allocation is Pareto optimal, i.e., there is no other feasible allocation such that at least one consumer is strictly better off while the others are not worse off. Suppose a CE allocation (x, y) is not balanced, i.e.,

$$\sum_{i=1}^n x_i \leq \sum_{j=1}^J y_j + \sum_{i=1}^n w_i.$$

Note that in Dr. Tian's note, two vectors $x \leq y$ means $x \leq y$ but $x \neq y$. Thus, at least for one good l ,

$$\sum_{i=1}^n x_i^l < \sum_{j=1}^J y_j^l + \sum_{i=1}^n w_i^l.$$

We can increase the consumption of one consumer, say 1, while keep other consumers get the same level, and make the above inequality hold with equality. By doing this, consumer 1 is strictly better off (for preferences are monotonic for all consumers) while the others are the same as before. This means the allocation (x, y) is Pareto optimal, a contradiction with the first welfare theorem.

In fact, from the proof we claim that **if each consumer have locally non-satiated preferences**, and **each commodity is strictly increasing in at least one consumer's preferences**, the CE allocation (if exists) should be balanced.

We can also prove the above claim by Walras' law. Locally non-satiated preferences implies Walras' law holds. Each commodity is strictly increasing in at least one consumer's preferences means if CE exists, then CE price should be strictly positive. Market clearing condition requires that $\sum_{i=1}^n x_i \leq \sum_{j=1}^J y_j + \sum_{i=1}^n w_i$. If $\sum_{i=1}^n x_i < \sum_{j=1}^J y_j + \sum_{i=1}^n w_i$, then we cannot have Walras' law, i.e., $p \cdot [\sum_{i=1}^n x_i - (\sum_{j=1}^J y_j + \sum_{i=1}^n w_i)] = 0$ does not hold.